

Dynamism brief: results review and validations in progress

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This is a short read-along of the business-dynamism work on the **business-dynamism** branch. Three new analyses since the last PDF, plus a hypothesis the user posed (“nearly the entire drop in registration is due to decrease in White-founded firms”) tested against our data.

Plot 1: US firms in existence + entry rate, 1978–2023

What this means for our coverage question. The dynamism decline is a *rate* story, not an *absolute count* story. There are more firms than ever, but the share of them that are new each year fell roughly in half. The denominator in our USPTO/BFS coverage ratio (BFS-HBA) is approximately constant in absolute counts 2004–2018 ($\sim 1.2\text{M}$ HBA/year), then jumps with the pandemic-era surge ($\sim 1.7\text{M}$ HBA/year, 2020–2024).

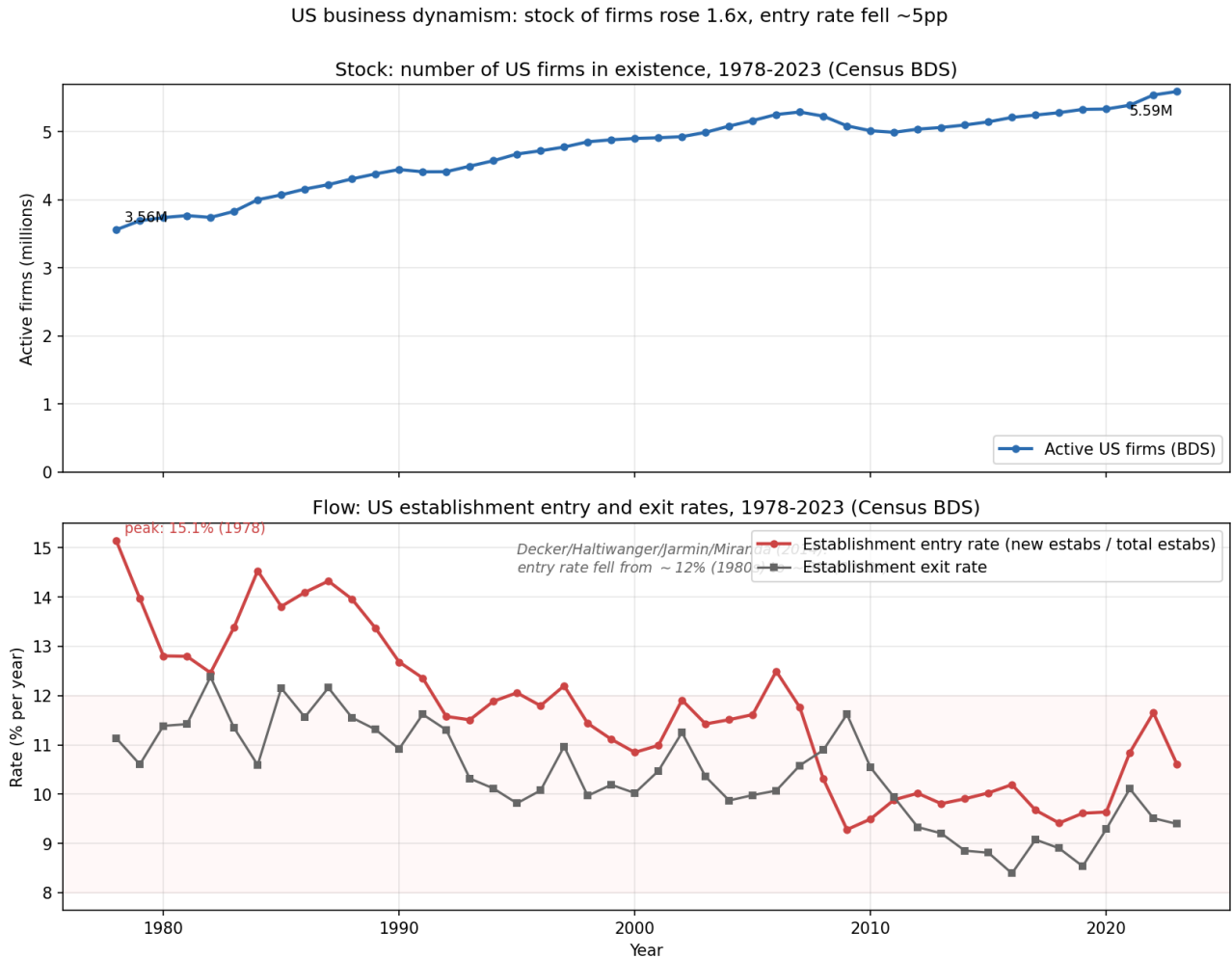


Figure 1: Top: the US firm stock (Census BDS, annual). 3.6M firms in 1978, 5.6M in 2023 – a 1.6 $\times$ rise. The denominator in the dynamism literature *grew*. **Bottom:** establishment entry and exit rates. The Decker, Haltiwanger, Jarmin, & Miranda (2014, *JEP*) “12% to 8%” decline is the entry-rate line. The exit rate fell in parallel; the gap between entry and exit (net firm change) compressed but stayed positive.

Plot 2: USPTO debut filings by surname-imputed ethnicity, absolute counts

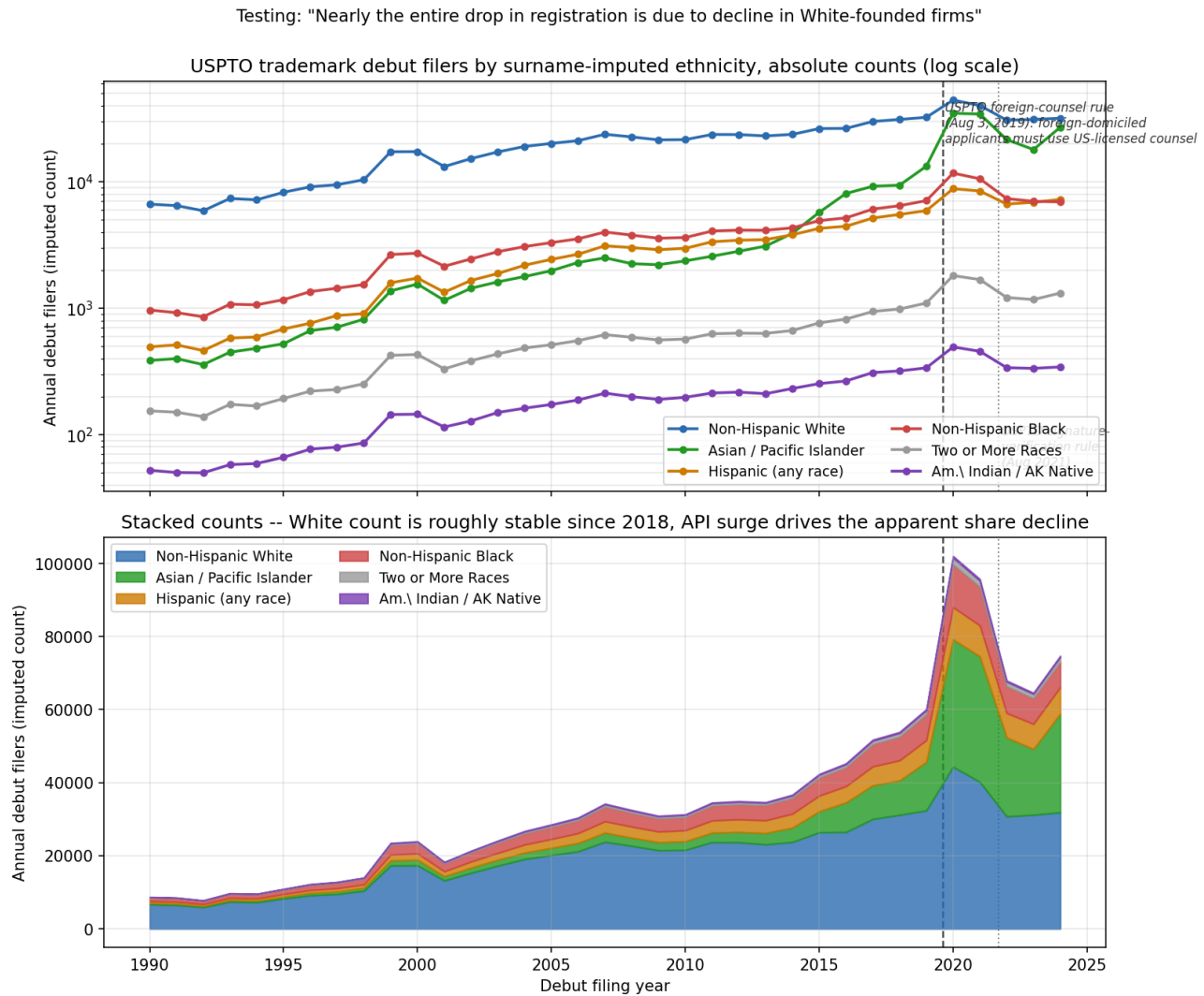


Figure 2: Top: log-scale absolute counts of USPTO debut filers whose name parses as a personal name, by surname-imputed race/ethnicity (Census 2010 surname probabilities). **Bottom:** stacked-area linear-scale of the same. Dashed vertical line at 2019.6: USPTO foreign-counsel rule (Aug 3, 2019, requiring foreign-domiciled applicants to use US-licensed counsel). Dotted line at 2021.7: USPTO signature-verification rule.

Testing the hypothesis that “the entire drop in registration is due to decline in White-founded firms”. The data do not support this. **White debut count is not declining.** Absolute debut counts among White-classified surname filers were $\sim 31\text{K}$ in 2018, $\sim 32\text{K}$ in 2019, $\sim 30\text{K}$ in 2022, $\sim 32\text{K}$ in 2024 – essentially flat over a seven-year window. The pandemic years (2020–2021) spiked to 40–44K, then reverted.

What did change is Asian/Pacific Islander. The API-classified count went from $\sim 9\text{K}$ in 2018 to 35K in 2020, then to $\sim 22\text{K}$ in 2022, then to $\sim 27\text{K}$ in 2024. The Aug-2019 foreign-counsel rule and the Aug-2021 signature-verification rule are both visible: each is followed by an API retreat (foreign filers withdrew under the new compliance burden), but the retreat is partial, and API counts in 2024

remain $\sim 3\times$ their pre-2018 level.

The fall in White *share* (75% in 1990s, 44% in 2020–24) is therefore an arithmetic consequence of API *growth*, not a White *decline*. The dynamism decline (12% $\rightarrow$ 8% in BDS) is about firms in the BLS/Census universe, not in our trademark sub-population, so this finding does not refute the dynamism story itself – it refutes *a particular causal attribution* of the trademark-side shift to White-founded firms specifically.

Plot 3: Sectoral coverage (NICE–NAICS), 2004–2024

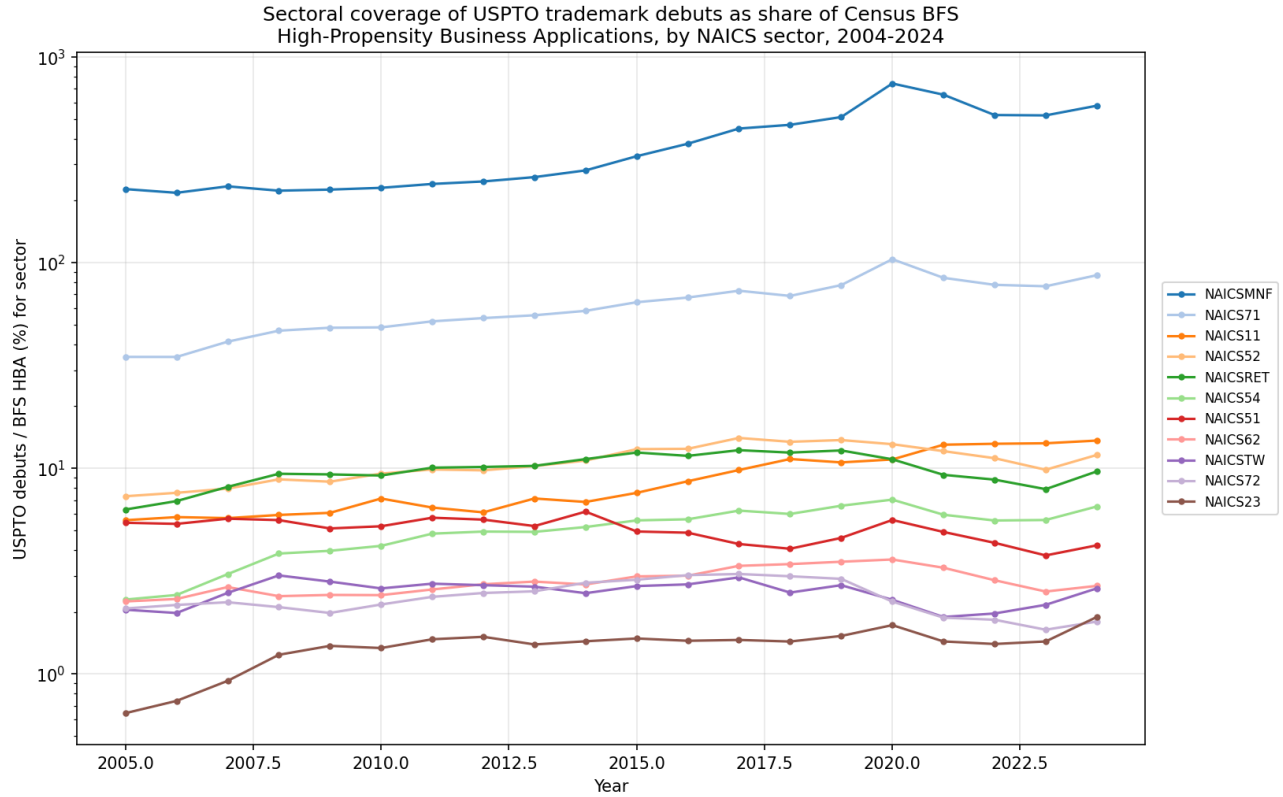


Figure 3: USPTO debut filings as a share of BFS High-Propensity Business Applications, by NAICS sector, log scale. Sectoral crosswalk is the first-cut hand-built mapping in `reference_data/nice_naics_crosswalk.csv`.

Coarseness of the first-pass crosswalk. Manufacturing (NAICSMNF) sits implausibly high ($\sim 588\%$ in 2019–2024) because the hand-built crosswalk assigns nearly every NICE goods class (001–034) to Manufacturing, including NICE 009 which alone has $\sim 3\text{M}$ filings and is dominantly *software/tech* businesses, not hardware manufacturing. Hall, Helmers, Rogers, & Sena (2014, *JEL*) document this exact NICE–NAICS attribution problem and recommend allocating 009 across NAICS 334 (electronics manufacturing), 511 (publishing), and 541 (professional services) via goods/services text similarity rather than a single primary code. This refinement is a clear next step.

The remaining sectors fall in plausible ranges:

- Arts/Entertainment (71): 85% – artists/musicians/studios trademark heavily
- Agriculture (11): 12% – farm trademark filings against ag SS-4 base
- Finance/Insurance (52): 12%
- Retail Trade (44–45): 10%
- Prof. Sci. Tech. (54): 6%
- Information (51): 5% (artificially low because software pulled into MNF)
- Health Care (62): 3%
- Transport/Warehouse: 2%

- Accommodation/Food (72): 2% – restaurants trademark rarely
- Construction (23): 2% – contractors rarely trademark

The Accommodation/Food number (2%) is worth flagging: this is where “IHOP” / franchise consolidation lives, and the very-low coverage means trademark data is a poor lens on dynamism in this sector. We would need state Secretary-of-State filings or SBA loan data to see independent-restaurant dynamics.

Coverage of new business formation (recap, with refinement)

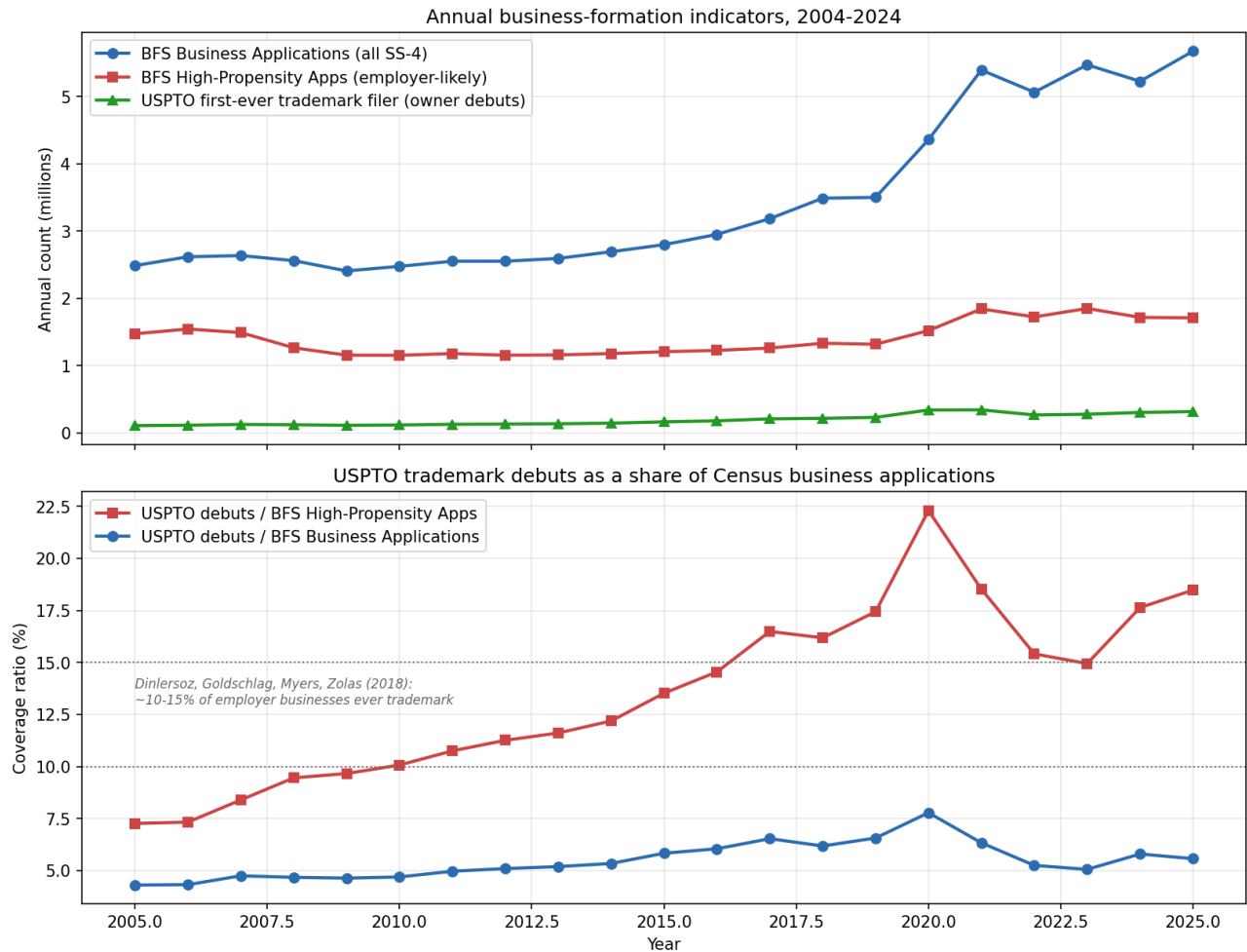


Figure 4: From the earlier write-up: USPTO debut filers vs. BFS BA and HBA, 2004–2024. Headline: 8% in 2004–2008 rising to 17.6% in 2019–2024 against the HBA denominator.

The two-decade rise in coverage is real and substantial, but the mechanism is now decomposable across the three readings we outlined:

1. Sectoral tilt toward IP-intensive sectors (more software / DTC brands relative to construction / restaurants): visible in the sectoral plot above, but not cleanly quantifiable until the NICE-NAICS crosswalk is refined.
2. Falling cost of filing: not separately testable from our data alone.
3. Denominator dilution by non-operating SS-4 filings: BFS-HBA specifically excludes these by design, so this reading is partly ruled out, but the 5%–6% ratio against BFS-BA (the unfiltered SS-4 series) is consistent with substantial non-operating dilution in the BA series.

Status of the foreign-applicant deconfounding

The cleanest test of whether the post-2018 API surge is foreign or US Asian-American requires extracting the <case-file-owner>/<state> and <case-file-owner>/<country> fields from the USPTO

XML.

Parser extension: done and verified. The streaming parser (`src/novelty/parse.py`) now extracts `owner_state` and `owner_country`. A smoke test on the 2017-05-21 daily file (1,558 records) confirms the fields populate correctly:

- 64.2% of records have a US state (US-domiciled).
- 35.6% of records have a country tag (foreign-domiciled).
- Top foreign country in the 2017 sample is already China (97 filings, $\sim 6.2\%$ of all 2017-05-21 applications).

Full re-parse: blocked on USPTO_ODP_API_KEY. The Open Data Portal at `data.uspto.gov` now gates bulk archive downloads behind an API key. To run the full backfile re-parse (`make -f Makefile.dynamism reparse`, expected runtime 30–60 minutes), add a key to `.env`:

`USPTO_ODP_API_KEY=<your-key-from-data.uspto.gov>`

Pipeline status (`Makefile.dynamism`)

Stage	Target	Status
Download reference data	<code>make ... download</code>	complete (BFS, Census surnames, BDS)
Hand-built crosswalk	<code>nice_naics_crosswalk.csv</code>	v1 in repo; refinement needed
Analysis (5 figures)	<code>make ... analysis</code>	complete
Re-parse USPTO XML	<code>make ... reparse</code>	blocked: API key
Review PDF	<code>make ... review</code>	this document
Open locally	<code>make ... open</code>	macOS open command

What I'd want feedback on

1. Confirm that the rejection of the White-decline hypothesis is read correctly: White absolute count is stable at $\sim 31\text{K}/\text{yr}$ since 2018; the share fall is API growth, not White decline.
2. Should I refine the NICE-NAICS crosswalk for NICE 009 specifically (split across MNF/Information/Prof. Sci. Tech. by goods/services text)? It would meaningfully change the sectoral plot but is non-trivial work.
3. Provision an `USPTO_ODP_API_KEY` so the full re-parse can run, or use the iCloud-cached daily files for a smaller sample?